

Transferability Across Different Molecular Systems and Levels of Theory with the Data-Driven Coupled-Cluster Scheme

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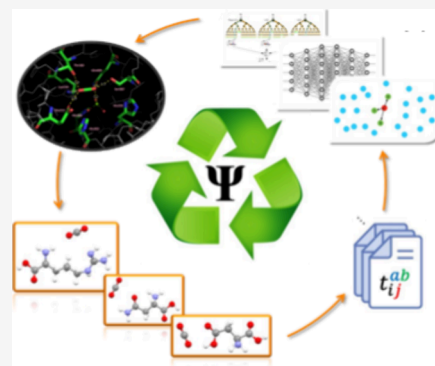
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ABSTRACT: Machine learning has recently been introduced into the arsenal of tools that are available to computational chemists. In the past few years, we have seen an increase in the applicability of these tools on a plethora of applications, including the automated exploration of a large fraction of the chemical space, the reduction of repetitive computational tasks, the detection of outliers on large databases, and the acceleration of molecular simulations. An attractive application of machine learning in molecular electronic structure theory is the “recycling” of molecular wave functions for faster and more accurate completion of complex quantum chemical calculations. Along these lines, we have developed hybrid quantum chemical/machine learning workflows that utilize information from low-level wave functions for the accurate prediction of higher-level wave functions. The *data-driven coupled-cluster* (DDCC) family of methods is discussed in this article together with the importance of the inclusion of physical properties in such hybrid workflows. After a short introduction to the philosophy and the capabilities of DDCC, we present our recent progress in extending its applicability to larger and more complex molecular structures and data sets. A significant advantage offered by DDCC is its transferability, with respect to different molecular systems and different excitation levels. As we show here, predicted wave functions at the coupled-cluster singles and doubles level of theory can be used for the accurate prediction of the perturbative triples of the CCSD(T) scheme. We conclude with some personal considerations with respect to future directions related to the development of the next generation of such hybrid quantum chemical/machine learning models.



INTRODUCTION

Artificial intelligence (AI) has permeated various domains of human activity, fundamentally transforming modern life in areas such as decision-making, strategic planning, and operational efficiency.¹ Numerous sectors, including business, finance, education, healthcare, and manufacturing, have integrated AI technologies into their daily operations. Notable applications of AI encompass natural language processing (NLP), computer vision, robotics, and machine learning (ML).² Machine learning, a subset of AI, enables computers to analyze and learn from data, improving their performance as they process larger data sets. Prominent examples of ML applications include predictive analytics, fraud detection, and recommendation systems.^{3–7}

AI and ML have also played pivotal roles in the past few years in sciences and engineering. Digital technologies coupled with advanced automation for data collection, analysis, curation, and prediction-making accelerate process optimization and scientific innovations.⁸ One of the key strengths of ML lies in its integration with methodologies and techniques that leverage extensive data sets. In such cases, ML can produce predictions with significantly reduced resources compared with traditional analytical methods. For chemistry in particular, where the chemical space is by default an

enormous database that is almost impossible to be extensively and systematically explored with traditional techniques, AI/ML have led to significant breakthroughs including drug design and development,⁹ chemical synthesis optimization,^{10–12} spectroscopy,^{13,14} analytical chemistry,¹⁵ chemoinformatics,^{16,17} predictive modeling,^{18–22} and computational chemistry.^{23–30}

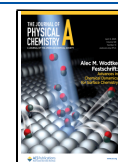
The strengths of ML have also been evident in applications related to quantum chemistry and, in particular, to molecular electronic structure theory,³¹ which is the focus of this perspective article. In molecular electronic structure theory, the molecular time-independent Schrödinger equation (within the Born–Oppenheimer approximation) is solved for the computation of eigenvalues (energies) and eigenstates (obtained from the molecular wave function). By solving the electronic problem for different nuclear positions, we obtain the potential energy surface of molecules and explore their

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properties at both equilibrium and far from equilibrium geometries, as well as for their ground and excited electronic states. However, the correlated motion of the electrons introduces a many-body problem that makes the exact solution of the time-independent Schrödinger equation for atoms and molecules with more than one electron impossible.³² Thus, the field of electronic structure theory examines the application of approximative models that systematically lead us to the exact solution of the Born–Oppenheimer Schrödinger equation. The Hartree–Fock (HF) theory is typically the starting point for more complex methods such as configuration interaction (CI), perturbation theory (PT), and coupled-cluster (CC) theory, which are commonly termed as post-HF methods.³³ The electronic structure theory models that arise from post-HF methods require the application of computational techniques with repetitive steps, such as the self-consistent field method or iterative solvers. These steps often have a demanding scaling (e.g., N^6 per iteration of coupled-cluster singles-and-doubles or CCSD, where N is the number of basis functions), and the slow convergence with respect to the number of iterations of these methods becomes a computational bottleneck for their application to larger molecular systems. Thus, ML becomes an attractive alternative methodology that can expedite the completion of such computational steps since it can perform repetitive tasks faster and more efficiently. A characteristic example is chatbots that are capable of handling routine customer service inquiries, allowing human agents to focus on complex issues that require more nuanced understanding and decision-making.^{34–36} Additionally, a large amount of data is generated during the performance of quantum calculations (e.g., integrals, molecular orbitals, intermediates), which contain directly or indirectly important information related to the molecular system under study.^{28,37} Again, ML can harness these data in a systematic and effective manner.

These ideas have been explored in the past years by numerous research groups for the development of exact or approximate electronic structure schemes, spanning from ML-based selective CI^{38,39} and CI solvers⁴⁰ which can also tackle variational quantum Monte Carlo^{41–43} or matrix product state Ansätze,⁴⁴ or Δ -ML approaches⁴⁵ that map low-order representations to learn energies or wave functions.^{46–52} Since the scope of this perspective article is not to review the recent literature, we refer the reader to a recent comprehensive review article³⁰ that covers in detail these and other similar ML-embedded quantum chemical schemes. On the contrary, the focus of this work is to provide an overview of an alternative but promising data-driven scheme that utilizes information from molecular wave functions. In the next sections, we will introduce the importance of input vectors that are used in ML models and how those are used effectively to “learn” either electronic structure energy terms or electronic wave functions.

■ QUANTUM CHEMICAL MOLECULAR REPRESENTATIONS – FROM DIMENSIONALITY REDUCTION TO MEANINGFUL PHYSICAL INFORMATION

A common theme in ML applications in chemistry is how we introduce relevant chemical information to the learner. In other words, how we convert or *represent* molecular structures and their properties in a manner that it can be read by a computer algorithm, since the common chemical nomencla-

ture (e.g., molecular or structural formula) is quite abstract or contains chemical intuition, to some extent. We also need to keep in mind that it is easier to provide as input to a computer program a vector that contains numbers or characters. For example, the Cartesian coordinates of each atom position can form an input vector which contains indirect information about the atom distances, but it does not hold any other geometric, electronic structure, or functionality-related information. Another important limitation is that the dimension of these vectors varies with system size, which can cause instabilities in the learning process. Thus, effective dimensionality reduction and the use of fixed-size input vectors to describe chemical structure and functionality are crucial for efficient ML. Thus, an active field of ML-related research is the vectorization of chemically relevant information for optimization of the learning process and performance of ML models.^{26,53–55}

We recently categorized these *molecular representations* into three different groups based on their aim and the information that is encoded.⁵⁵ The first category is the *graph-based representations* that consider every molecular structure is a graph by considering atoms as nodes and bonds as edges. The second category is the *topology-based representations* that extract additional information from the molecular graph by applying robust computational tools from algebraic topology and, in particular, from persistent homology.^{21,56} The third category is *physics-based representations* that contain information directly or indirectly from the physical properties of the molecule under consideration. Indicative examples of features included in physics-based representations are atomistic information, two-body, three-body or higher-order information, orbital information, local or total densities, and more. Very often, combinations of the aforementioned descriptors are used for the successful training of ML models (e.g., graph-based representations augmented by physics-based information).

From a physics perspective, the molecular wave function is the ultimate representation of a molecule. It contains all the geometric and electronic structure information, and by applying the corresponding quantum operators, we can retrieve its energy and properties. However, it is also becoming intractable as the system size increases so it cannot be used *per se* as input for ML surrogate models. For that reason, alternative physics-based representations that still hold relevant electronic structure information are required for ML-boosted quantum chemistry. Those should follow principles similar to those of the molecular representations discussed in the previous paragraph. Two representative examples are the work of Schütt et al., which uses atomic orbital and pairwise expansions to represent the molecular electronic structure of small molecules,^{46,57} and Welborn et al., who decomposed information that is included in (localized) molecular orbitals.⁴⁹ In principle, these N -electron models that generate such an orbital representation must be orbital-invariant.

■ LEARNING OBJECTIVES OF CHEMICAL MACHINE LEARNING

The majority of the models that we have discussed thus far are predicated on supervised learning, necessitating a data set of labeled annotations for training the workflows. Labeled data refer to data that are carefully annotated with tags (labels). These labels are used as “learnable output” during model optimization. In the context of image recognition, a datum is labeled to describe the image content (e.g., cat, hot dog, palm

tree, etc.). Upon exposure to millions of such images, the model undergoes optimization, eventually learning the image patterns (e.g., shape, color, and other characteristics) associated with specific outputs. In the domain of chemical ML, the learnable output can include molecular properties (e.g., biological activity of a drug), structural features (e.g., presence of a specific functional group), or, in the case of quantum chemistry ML models, energy terms, wave function parameters, or even molecular Hamiltonian terms. A short description of such different learnable levels is provided in the next paragraphs.

Computational chemistry heavily relies on the calculation of the relative energy differences (ΔE). For example, the enthalpy of a reaction (ΔH_{rxn}) is computed as the difference between the enthalpies of the products and the reactants. The accuracy of relative energy differences often relies on fortuitous error cancellation when truncated one- and N -electron expansions are used, and thus, they do not necessarily require the computation of highly accurate energies for each individual calculation. For that reason, relative energies for a plethora of different chemical applications have been a common learnable output for many chemical ML models, such as reaction barriers,^{22,58–60} enantioselectivity,⁶¹ spin ladders,⁶² and intermolecular interactions.^{21,63}

The actual challenge of ML applications in quantum chemistry is the development of models that can accurately provide the total energy of a molecular structure since, from that point, it becomes more straightforward to obtain full potential energy surfaces and molecular properties. Such models require additional sophistication in their design with respect to models that aim to predict chemical properties, but they typically offer enhanced transferability across different systems, especially when they satisfy fundamental physical properties. The most common examples are the ML force fields (MLFFs) that aim to capture the interatomic interactions and learn indirectly the physics of a molecular system or a material via a large set of highly flexible functions that are fitted to quantum chemical data.^{25,30} Usually, the total energy E_{tot} is computed as a summation of all atom energies, which is achievable since the model is able to learn the interaction of each atom with their neighboring atoms:

$$E_{\text{tot}} = \sum_i^{N_{\text{atoms}}} E_i[\mathcal{N}(i)] \quad (1)$$

where $E_i[\mathcal{N}(i)]$ is the energy of atom i in a functional form since it depends on all the neighboring atoms. MLFFs and ML-generated potentials have been used in many different molecular applications ranging from chemical reactivity⁶⁴ to excited states⁶⁵ and molecular properties.

An alternative approach was introduced by Miller and co-workers, who developed sophisticated ML models that aim to “learn” the electron pair energy e_{ij} .^{49,50} By invoking the Nesbet theorem,⁶⁶ they were able to recover the total correlation energy $E_{\text{tot}}^{\text{corr}}$ of a molecule from the individual pair energies:

$$E_{\text{tot}}^{\text{corr}} = \sum_{i < j}^{N_{\text{electrons}}} e_{ij} \quad (2)$$

In their models, the target correlation energy is at the coupled-cluster singles-and-doubles (CCSD) level or coupled-cluster singles-and-doubles with perturbative triples [CCSD(T)] level, while the input vector is based on electronic structure features

from the computationally cheaper Hartree–Fock theory. The feature space consists of orbital energies, molecular orbital coefficients, and Fock, Coulomb, and exchange matrix elements. This can be shown schematically as:

$$\Psi_{\text{HF}}/\Psi_{\text{MP2}} \xrightarrow{\text{ML}} E_{\text{CCSD/CCSD(T)}}^{\text{corr}} \quad (3)$$

Inspired by this work, we have developed a family of ML methods under the name *data-driven coupled-cluster* (DDCC). In DDCC, instead of learning the CCSD/CCSD(T) correlation energy, our focus is on learning the underlying parameters (i.e., CCSD t_2 amplitudes) of the targeted electron correlation method:

$$\Psi_{\text{HF}}/\Psi_{\text{MP2}} \xrightarrow{\text{DDCC}} \Psi_{\text{CCSD/CCSD(T)}} \quad (3a)$$

In the next sections, we will present the design principles of DDCC, its successes, solutions for its limitations, and the future steps that our research group has started implementing to these ML schemes.

THE DATA-DRIVEN COUPLED-CLUSTER SCHEME

The coupled-cluster theory utilizes an exponential Ansatz that aims to capture the missing electron correlation from HF theory by considering electronic excitations to virtual orbitals (unoccupied orbitals) from occupied orbitals:

$$|\Psi_{\text{CC}}\rangle = \exp(\hat{T})|\Psi_0\rangle \quad (4)$$

In our work, the canonical HF reference state is considered. The cluster operator $\hat{T}$ promotes (or “excites”) electrons from the zeroth-order HF wave function Ψ_0 and, in principle, considers all possible electron excitation combinations (one-electron excitations, two-electron excitations, three-electron excitations, etc.). For practical applications, a truncated expansion is used since the calculation of higher-order excitations is computationally demanding. For example, CCSD considers only single and double electron excitations through the one- and two-electron excitation operators, respectively:

$$\hat{T}_1 = \sum_{i,a} t_i^a a_a^\dagger a_i \quad (5)$$

$$\hat{T}_2 = \frac{1}{4} \sum_{i,j,a,b} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i \quad (6)$$

where a_i and a_j are annihilation operators that remove an electron from the occupied orbitals i and j , respectively, and $a_a^\dagger$ and $a_b^\dagger$ are creation operators that place the electrons at the unoccupied orbitals a and b , respectively. The t_i^a and t_{ij}^{ab} are the one- and two-electron amplitudes, respectively, that need to be determined. These amplitudes constitute the parameters of the CCSD wave function $\Psi_{\text{CC}}(t_1, t_2)$, where t_1 and t_2 represent the full manifold of all possible one- and two-electron amplitudes of a given molecule. A variational solution to the truncated expansion leads to exponential scaling, and it can only be applied to very small molecules,^{67–69} so in general it is avoided for a more compact scheme through the solution of a set of projected equations:

$$\langle \Psi_0 | \exp(-\hat{T}) \hat{H} \exp(\hat{T}) | \Psi_0 \rangle = E_{\text{CC}} \quad (7)$$

$$\langle \Psi_i^a | \exp(-\hat{T}) \hat{H} \exp(\hat{T}) | \Psi_0 \rangle = 0 \quad (8)$$

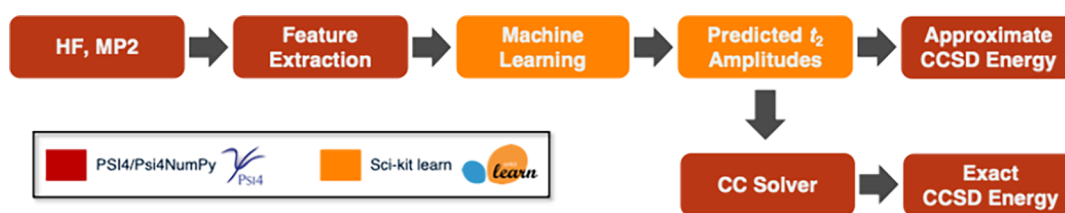


Figure 1. Schematic diagram of the data-driven coupled-cluster scheme. Steps that involve electronic structure theory software are shown in red boxes; steps performed by machine learning libraries are shown in orange boxes.

$$\langle \Psi_{ij}^{ab} | \exp(-\hat{T}) \hat{H} \exp(\hat{T}) | \Psi_0 \rangle = 0 \quad (9)$$

The solution of eq 7 leads to the coupled-cluster energy expression:

$$E(t_1, t_2) = \sum_{i < j, a < b} \langle ij || ab \rangle t_{ij}^{ab} + \sum_{i < j, a < b} \langle ij || ab \rangle t_i^a t_j^b \quad (10)$$

where $\langle ij || ab \rangle$ is the standard two-electron (four-index) integral. The solution of the full set of nonlinear equations of eqs 8–9 provides the t_1 and t_2 amplitudes. This requires an iterative solver, and this process scales as N^6 per iteration. In conventional CC implementations, the initial values of the t_2 amplitudes are obtained from the two-electron amplitudes of the computationally less demanding MP2 method (single step with N^5 scaling):

$$t_{ij,[MP2]}^{ab} = \frac{\langle ij || ab \rangle}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b} \quad (11)$$

while the one-electron amplitudes t_1 are set to zero. In eq 11, the ε_p terms of the denominator are the orbital energies from HF theory ($p = ij, a, b$).

We can consider that the CC iterative solver is a systematic mathematical process that begins from the $t_{2[MP2]}$ amplitudes until it converges to the $t_{2[CCSD]}$ amplitudes of CCSD. Depending on the number of amplitudes that need to be determined as dictated by the molecular size and structure, as well as the basis set size, or the near-degeneracies of the electronic states, this process might converge after a few (10–12 iterations) or more (50+ iterations), while for difficult cases it can even show oscillatory behavior. Since this is a repetitive process, the application of ML methods can expedite the convergence of the iterative solver. Based on that, we made a hypothesis that the information that is stored in the two-electron excitation $ij \rightarrow ab$ at the HF and MP2 level can be used to construct a descriptor-based vector that can be introduced as input to ML models. Such a model can predict a new t_{ij}^{ab} amplitude that is closer to the exact CC amplitude. That hypothesis was the foundation for the development of the DDCCSD scheme, which aims to recover the CCSD energy at MP2 cost.

The basic steps of the DDCCSD methodology are schematically shown in Figure 1. All initial features that represent a two-electron excitation $ij \rightarrow ab$ are obtained from the HF and MP2 levels. We have developed a 30-dimensional feature vector that decomposes the t_{ij}^{ab} amplitude into electronic structure terms that describe the properties of the orbitals i, j, a , and b . These characteristic features are the MP2 amplitude itself that implicitly depends on the HF reference wave function, and the orbital energies ε_p which are further decomposed into one-electron Hamiltonian, Coulombic, and exchange components.⁷⁰ Once those are collected, they are

introduced into the trained ML model. So far, we have explored different learners, including k-nearest neighbors (kNN),⁵² random forest (RF),⁷¹ and neural networks (NN). A properly trained learner should be able to predict a two-electron amplitude, $t_{ij,[DDCC]}^{ab}$, that is close to the exact CCSD t_{ij}^{ab} amplitude. At this point, we can choose between two different paths. The first is to perform a single CCSD iteration step that will update only the t_1 amplitudes (which were initially set to zero), without modifying the $t_{2[DDCC]}$ amplitudes. The two sets of $t_{1[DDCC]}$ and $t_{2[DDCC]}$ amplitudes are introduced into eq 10, and an approximate CCSD energy is computed (Figure 1, top right path). The alternative path is to introduce the $t_{2[DDCC]}$ amplitudes into the CC solver, run it until convergence, and thus, recover the exact CCSD energy (Figure 1, bottom path). Since the predicted $t_{2[DDCC]}$ amplitudes are close to the final ones, the CC solver converges in less iterations than when we start from the $t_{2[MP2]}$ amplitudes. In our initial implementation, we utilized the Psi4NumPy package⁷² for the HF/MP2 calculations, as well as for collecting CCSD data (amplitudes and total correlation energies) for the training of ML models (either from Scikit-learn⁷³ or PyTorch).

In a follow-up article,⁷⁴ we explored the local nature of electron correlation on ML-driven methodologies. A key outcome of that study was the transferability that was achieved by combining localized orbitals together with powerful ML algorithms. More recently, we introduced localized orbitals into the original DDCCSD scheme, and we achieved a 10-fold increase in accuracy, independently if we used a kNN or RF model.⁷¹

Another limitation of the initial DDCCSD implementation was the large number of data (30-dimensional feature vector, t_2 amplitudes) that were generated as the system size was increasing. This is an outcome of the increasing number of two-electron excitations as we consider larger molecules (more correlated electrons) or of the increasing basis set size (more virtual orbitals). In addition, the majority of these amplitudes have values very close to zero (absolute values that are below 10^{-6}), and less than 1% of the total number of data used for DDCCSD model training spans over values that provide a substantial contribution to the CCSD correlation energy. In other words, the amplitude space explored by DDCC is biased toward the close-to-zero values. To address these two issues, we developed five sampling methods that allow a more balanced selection of the full amplitude space which were grouped based on their design principles (*data-driven* or *physics-driven*).⁷¹ The data-driven schemes were based on scoring functions, *k*-means clustering, or probabilistic weights, and helped to collect a more balanced amplitude space for DDCC training. The physics-driven schemes considered the amplitude $t_{ij,[MP2]}^{ab}$ or the correlation energy values e_{ij}^{ab} of each $ij \rightarrow ab$ excitation. A cutoff value was introduced for $t_{ij,[MP2]}^{ab}$ or

e_{ij}^{ab} , which helped us select data points used for the training, testing, and validation of the model. These cutoff values can be considered as an additional hyperparameter of the DDCC scheme that can be modified and optimized by the user. Overall, it was found that all five selection schemes require less data to achieve similar accuracy as models trained on the full amplitude data set. We found that the probabilistic selection was the most consistent since it provided faster convergence to the full models as the number of data is increasing. On the contrary, the physics-driven schemes were able to provide higher accuracy than the full model with only 15–30% of the full amplitude data set, which was accredited to a more balanced sampling of the full amplitude space.⁷⁰

Model transferability between molecules of different size was demonstrated in our prior work.⁷¹ DDCCSD models trained on data from five organic molecules (diethyl ether, divinyl ether, vinyl alcohol, methyl vinyl ether, and allyl alcohol) performed equally well with respect to models trained on individual molecules (MAEs of 1.04 and 0.69 from models trained on mixed data and individual molecules, respectively).

We also mention a few studies that share similarities with the DDCC scheme. Ng, Liang, and Yang⁷⁵ developed an alternative featurization based on localization molecular orbitals that utilizes orbital-specific virtuals (OSVs), which is further coupled to neural networks. Schrader and Kvaal⁷⁶ used Gaussian processes for generating improved initial guess for the coupled cluster amplitudes. Peyton et al. developed a density tensor representation for the estimation of molecular properties at the coupled-cluster level.⁷⁷ Ruth et al. developed a delta-ML scheme that predicts CCSD(T) energies from HF, MP2, and CCSD data for accurate thermochemistry.⁷⁸ Finally, along the same principles, we have developed a scheme that utilizes a physics-based feature vector for estimating the error of variational two-electron reduced-density matrix (v2RDM) from full configuration interaction (FCI).⁷⁹

TRANSFERABILITY ACROSS MOLECULES AND EXCITATION LEVELS

The application of *machine learning in quantum chemistry* has seen significant advancements in recent years, driven by progress in data science and computational technologies. However, a critical factor for enhancing the accuracy and success of these methodologies lies in the integration of additional physical properties into ML workflows. As discussed previously, encoding wave function properties that capture electron correlation, such as pair energies or wave function parameters, is essential. The next pivotal challenge is to embed domain-specific physics directly into the machine learning models, moving beyond treating these algorithms as black boxes. Such *physics-informed machine learning* approaches⁸⁰ have recently gained considerable traction across various scientific and engineering disciplines.

Let us first examine the advantages of the current DDCC schemes by the introduction of the physics behind the two-electron correlation before we turn our attention to the physics-inspired ML models. The focus of DDCC is to capture the properties of the two-electron excitation $ij \rightarrow ab$ which, in a local basis, has a universal character across different molecular structures. This offers an intrinsic transferability across different atomistic environments, as we have previously demonstrated.^{52,71,74} For example, DDCC models trained on data from water monomer, dimer, and trimer geometries are able to accurately capture the CCSD correlation energy of

larger water clusters (hexamer and decahexamer water clusters).⁵²

We also used conformers from a set of organic molecules to compare the accuracy of the models trained with localized and canonical molecular orbitals.⁷¹ Here we were able to achieve a 10-fold increase in accuracy using the localized orbitals. In the same study, we were able to use data selection schemes to reduce the training data and achieve higher accuracies for the same set of organic molecules. This will improve the applicability of the models.

Another important advantage of “learning” the CCSD wave function parameters rather than the correlation energy (total or pair correlation energies) is the transferability between different levels of theory across the coupled-cluster expansion. Pair-electron theories often form the basis for higher-order excitation energy terms. For example, the well-established perturbative correction $E_{(T)}$ to the full triples excitation of coupled-cluster theory of the CCSD(T) methodology⁸¹ requires the computation of the triples amplitudes t_{ijk}^{abc} , which are generated from the already precomputed t_{ij}^{ab} amplitudes of CCSD. Based on that, we have introduced the predicted $t_{2[DDCC]}$ amplitudes of DDCC into the well-known equations of the fourth and fifth order components of CCSD(T). To test this hypothesis, we trained DDCC schemes with two different learners (kNN and RF) and three different basis sets (STO-3G, 6-31G, cc-pVDZ). We considered the symmetric stretch of the O–H bonds of water around its equilibrium geometry. For each model, we utilized data from 20 H₂O geometries, and we tested it on energies from 30 different H₂O geometries that were not part of the training set. Deviations from each of these models are shown in Table 1, while Figure 2 shows the

Table 1. DDCCSD and DD(T) Errors, Together with Percent DD(T) Deviations Computed by Eq. 12, as Obtained from Models Trained with the k-Nearest Neighbors (kNN) and Random Forest (RF) Learners^a

DDCCSD(T) – kNN			
	DDCCSD Error	DD(T) Error	Percent DD(T) deviation
STO-3G	1.552	0.004	3.9%
6-31G	0.826	0.081	2.9%
cc-pVDZ	0.928	0.140	2.3%
DDCCSD(T) – RF			
	DDCCSD Error	DD(T) Error	Percent DD(T) deviation
STO-3G	0.918	0.004	4.2%
6-31G	1.431	0.097	4.0%
cc-pVDZ	3.419	0.247	4.9%

^aHere we use a training set of 20 water molecule geometries and a test set of 30 water molecule geometries with STO-3G, 6-31G and cc-pVDZ basis sets.

potential energy curve from the conventional CCSD/CCSD(T) methods together with data-driven variants DDCCSD/DDCCSD(T), respectively. The percent deviation of $E_{(T)}$ from the DDCCSD(T) level, labeled here as DD(T), are shown on Table 1 and computed as:

$$\% \text{ DD(T) deviation} = \frac{\text{DD(T) Error}}{E_{(T)}} \times 100 \quad (12)$$

For Figure 2, we considered only the models trained with the RF algorithm, but similar results were obtained from the kNN models. Overall, we found that the DDCCSD and DDCCSD(T)

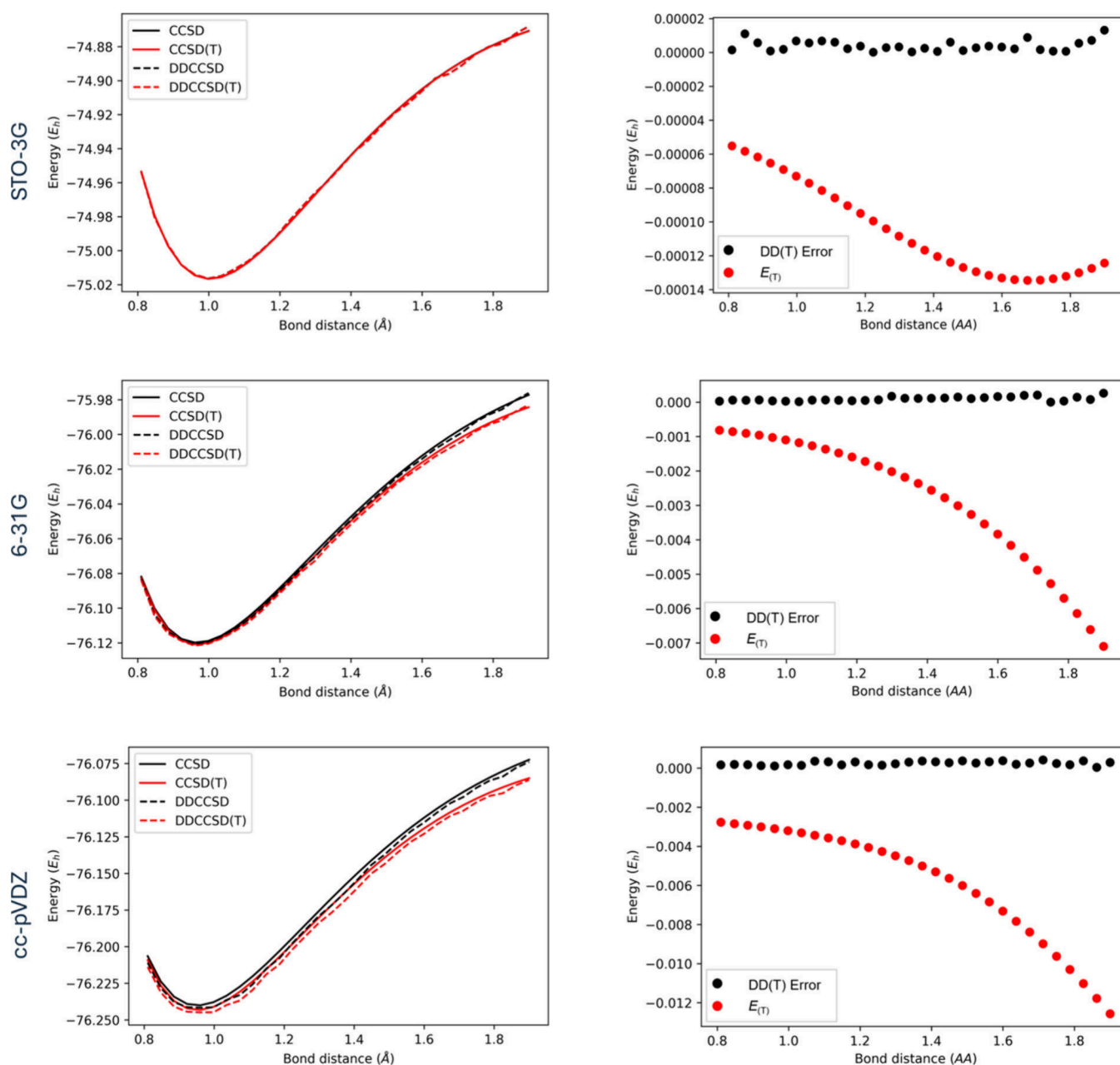


Figure 2. Potential energy curves (left) along the symmetric O–H stretch of the H_2O molecule, as computed by CCSD, CCSD(T), DDCCSD, and DDCCSD(T) with STO-3G, 6–31G, and cc-pVDZ basis sets. The E_{T} energies together with the deviations of $E_{\text{DD(T)}}$ along the O–H stretch are shown on the right. The DDCCSD/DDCCSD(T) models were trained with the random forest (RF) algorithm.

(T) curves are in close agreement with the parent CCSD and CCSD(T) methods, respectively, independent of the basis set choice. More interestingly, we found that even if the E_{T} energy term becomes more negative at distances far from the equilibrium geometry (red points on the right plots of Figure 2), the errors of DD(T) remain constantly around zero. Thus, we conclude that the $E_{\text{DD(T)}}$ errors are independent of the E_{T} magnitude, and their accuracy depends only on the quality of the t_2 amplitudes predicted by the DDCCSD.

To illustrate the transferability of DDCCSD(T), we trained a model with data generated by heterogeneous molecules and attempted to predict CCSD(T) energies from out-of-sample molecules. We considered 30 near-equilibrium water conformers where the O–H bond lengths were varied in the 0.8–1.2 Å range, and 30 near-equilibrium methane conformers

where the C–H bond lengths were varied in the same 0.8–1.2 Å range. All calculations were performed with the cc-pVDZ basis set, and the trained DDCCSD(T) scheme utilized a random forest. For the effective sampling of our data, we applied a $t_{\text{ij},[\text{MP2}]}$ amplitude cutoff of $1 \times 10^{-3.25}$, which reduced the input data to 22.75% of the total number of amplitudes. The model accuracy was evaluated by using 30 methanol conformers where the O–H bond lengths were varied in the 0.8–1.2 Å range. The total MAE of 1.066 mE_h is comparable to the errors reported on Table 1 and provides a roadmap for the development of robust DDCCSD(T) models with broader applicability.

To demonstrate the acceleration gained from DDCCSD methods and the capability of the method to be applied to open-shell systems, we performed calculations on the following

molecular systems: CH_3 (neutral radical), CH_4 , CH_3CH_3 , $\text{CH}_3\text{CH}_2\text{CH}_3$, CH_3CH_2 (neutral, radical), and CH_3CHCH_3 (neutral, radical), as well as the C–H dissociation curve of CH_4 , the C–C dissociation curve of CH_3CH_3 , and the potential energy curve of the hydrogen atom transfer from CH_4 to CH_3 . First, the DDCCSD models were trained with data from individual molecular systems (Table 2). We used 30

Table 2. Average Error Per Electron (in mE_h) for DDCC and DD(T)^a

System	Average DDCCSD error per electron	Average DD(T) error per electron	Average CCSD time	Average DDCCSD time
CH_3 radical	0.222	0.034	7.3	0.5
CH_4	0.211	0.040	16.2	1.3
CH_3CH_2 radical	0.401	0.078	213.4	15.0
CH_3CH_3	0.337	0.072	289.1	29.7
CH_3CHCH_3 radical	0.334	0.114	1930.3	174.4
$\text{CH}_3\text{CH}_2\text{CH}_3$	0.405	0.077	2265.5	255.5
CH_4 dissociation	0.231	0.036	13.2	1.2
CH_3CH_3 dissociation	0.258	0.045	279.1	29.0
$\text{CH}_4\text{--CH}_3$ hydrogen transfer	0.474	0.057	660.4	48.6
Average	0.319	0.061		

^aTimings (in seconds) are also reported for conventional CCSD and DDCCSD. All models were trained with data computed with the cc-pVDZ basis set.

conformers from each molecule for training and 30 different conformers for testing. The random forest algorithm and the LA amplitude selection scheme with a 1×10^{-4} threshold was applied in all DDCC models (both for training and testing) together with the cc-pVDZ basis set and canonical molecular orbitals. Overall, we found that DDCCSD and DD(T) have average deviations per electron of 0.319 and 0.061 mE_h from the conventional CCSD and (T), respectively, while DDCCSD has a 10-fold acceleration compared to CCSD (8.8 up to 14 times faster than CCSD).

We also considered transferability of the DDCCSD and DDCCSD(T) methods by training models with data from different molecular systems. In particular, we considered a training set consisted of five conformers each from CH_3 (neutral radical), CH_4 , CH_3CH_3 , CH_3CH_2 (neutral radical), and CH_3CHCH_3 (neutral radical), and tested on five conformers each from these systems that were not part of the training (Table 3). The DDCCSD and DD(T) average deviations per electron of 0.325 and 0.057 mE_h from the conventional CCSD and (T), respectively, are almost identical to the errors from the models trained on individual molecular systems (0.319 and 0.061 mE_h , respectively).

Finally, we would also like to mention that the energy expression of eq 10 holds for any coupled-cluster excitation rank, from CCSD up to full CC excitation expansion. Therefore, a DDCC scheme can be trained, in principle, with higher-order data (e.g., CCSDT or CCSDTQ) and provide energy estimations close to that excitation level, by solely learning the updated t_2 amplitudes from that particular level of theory.

Table 3. Average DDCCSD and DD(T) Errors per Electron (in mE_h) from the DDCCSD and DDCCSD(T) Models Trained on Mixed Data^a

Test molecule type	Average DDCCSD error per electron	Average DD(T) error per electron
CH_3 (radical)	0.255	0.014
CH_4	0.204	0.047
CH_3CH_2 (radical)	0.372	0.057
CH_3CH_3	0.354	0.087
CH_3CHCH_3 (radical)	0.440	0.082
Average	0.325	0.057

^aAll models trained with data computed with the cc-pVDZ basis set.

FUTURE DIRECTIONS: FROM DATA-DRIVEN METHODOLOGIES TO LARGE-SCALE, PHYSICS-INSPIRED MACHINE LEARNING

To conclude, recent advances in the DDCC method have helped us to extend the applicability of the method after providing solutions to specific limitations, such as the collection of more balanced training sets by introducing amplitude selection schemes. The mitigation from kNN learners to RF also helped us to develop more portable models, since we do not need to store enormous sets of data for predicting $t_{2[\text{DDCC}]}$ amplitudes that are required by instance-based learning models. Transferability between different molecular systems and different coupled-cluster excitation levels can also be achieved, as long as there are adequate data for training the DDCC models. The current DDCC models offer a significant speedup since they require input from an MP2 calculation but depend on data collected from prior calculations together with adequate training. We envision that DDCC schemes can give an important advantage to computational processes where repetitive coupled-cluster calculations are required, such as geometry optimizations or ab initio molecular dynamics (AIMD) simulations. Data are generated during the first computational steps, DDCC model training is performed on the fly, and the hybrid scheme significantly accelerates the next computation or simulation steps. Future directions should include the incorporation of more physical properties in DDCC, not only on the input vector, but primarily on the DDCC architecture itself. This can be achieved by introducing more flexible ML models that are based on neural networks, a direction that recently our group has started exploring. Some alternative methodologies that we envision to apply in the future in the field of ML in quantum chemistry are the introduction of generative and foundation models, more autonomous data generation via reinforcement learning, and operator learning.

A final note should be given, with respect to the broader applicability of such ML/quantum chemical methods. We strongly believe that these hybrid schemes can revolutionize in the years to come how we perform quantum chemical calculations, since they allow us to “recycle” prior information that is collected from successfully completed calculations. However, those methodologies can also be coupled with existing approaches that have been effectively applied in our field, such as embedding schemes, quantum mechanics/molecular mechanics (QM/MM) methodologies, fragmentation schemes, high-performance computing, basis set extrapolation and explicitly correlated schemes, and more.

Recent advances in AI/ML architectures have provided additional flexibility to users on the development of tools tailored to a specific purpose without treating ML as a black box. While in most cases ML is applicable to interpolate within the boundaries of the training sets, methodologies such as active learning, reinforcement learning, and generative models have shown the potential to elevate the applicability of AI/ML to unexplored data sets. In the next few years, we are expecting to see novel, hybrid models that couple these technologies with electronic structure theory. It is thus an interesting challenge for AI/ML to move toward the predictive power obtained from solutions of the Schrödinger equation from first-principles, which can be applied to completely unexplored species and phenomena.

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REFERENCES

- (1) Xu, Y.; Liu, X.; Cao, X.; Huang, C.; Liu, E.; Qian, S.; Liu, X.; Wu, Y.; Dong, F.; Qiu, C. W.; et al. Artificial intelligence: A powerful paradigm for scientific research. *Innovation* **2021**, 2 (4), No. 100179.
- (2) Zhang, C.; Lu, Y. Study on artificial intelligence: The state of the art and future prospects. *J. Ind. Inf. Integr.* **2021**, 23, 100224.
- (3) Zhang, Q.; Lu, J.; Jin, Y. Artificial intelligence in recommender systems. *Complex Intell. Systems* **2021**, 7 (1), 439–457.
- (4) Nti, I. K.; Adekoya, A. F.; Weyori, B. A.; Nyarko-Boateng, O. Applications of artificial intelligence in engineering and manufacturing: a systematic review. *J. Intell. Manuf.* **2022**, 33 (6), 1581–1601.
- (5) Xu, W.; Ouyang, F. The application of AI technologies in STEM education: a systematic review from 2011 to 2021. *Int. J. STEM Educ.* **2022**, 9 (1), No. 59.
- (6) Alowais, S. A.; Alghamdi, S. S.; Alsuhbany, N.; Alqahtani, T.; Alshaya, A. I.; Almohareb, S. N.; Aldairem, A.; Alrashed, M.; Bin Saleh, K.; Badreldin, H. A.; et al. Revolutionizing healthcare: the role of artificial intelligence in clinical practice. *BMC Med. Educ.* **2023**, 23 (1), 689.
- (7) Masciari, E.; Umair, A.; Ullah, M. H. A Systematic Literature Review on AI-Based Recommendation Systems and Their Ethical Considerations. *IEEE Access* **2024**, 12, 121223–121241.
- (8) Greenaway, R. L.; Jelfs, K. E.; Spivey, A. C.; Yaliraki, S. N. From alchemist to AI chemist. *Nat. Rev. Chem.* **2023**, 7 (8), 527–528.
- (9) Mullowney, M. W.; Duncan, K. R.; Elsayed, S. S.; Garg, N.; van der Hooft, J. J. J.; Martin, N. I.; Meijer, D.; Terlouw, B. R.; Biermann, F.; Blin, K.; et al. Artificial intelligence for natural product drug discovery. *Nat. Rev. Drug. Discovery* **2023**, 22 (11), 895–916.
- (10) de Almeida, A. F.; Moreira, R.; Rodrigues, T. Synthetic organic chemistry driven by artificial intelligence. *Nat. Rev. Chem.* **2019**, 3 (10), 589–604.
- (11) Burger, B.; Maffettone, P. M.; Gusev, V. V.; Aitchison, C. M.; Bai, Y.; Wang, X.; Li, X.; Alston, B. M.; Li, B.; Clowes, R.; et al. A mobile robotic chemist. *Nature* **2020**, 583 (7815), 237–241.
- (12) Leonov, A. I.; Hammer, A. J. S.; Lach, S.; Mehr, S. H. M.; Caramelli, D.; Angelone, D.; Khan, A.; O'Sullivan, S.; Craven, M.;

Wilbraham, L.; et al. An integrated self-optimizing programmable chemical synthesis and reaction engine. *Nat. Commun.* **2024**, *15* (1), 1240.

(13) Meza Ramirez, C. A.; Greenop, M.; Ashton, L.; Rehman, I. u. Applications of machine learning in spectroscopy. *Appl. Spectrosc. Rev.* **2021**, *56* (8–10), 733–763.

(14) Guo, S.; Popp, J.; Bocklitz, T. Chemometric analysis in Raman spectroscopy from experimental design to machine learning-based modeling. *Nat. Protoc.* **2021**, *16* (12), 5426–5459.

(15) Mayer, M.; Baeumner, A. J. A Megatrend Challenging Analytical Chemistry: Biosensor and Chemosensor Concepts Ready for the Internet of Things. *Chem. Rev.* **2019**, *119* (13), 7996–8027.

(16) Molga, K.; Szymkuc, S.; Grzybowski, B. A. Chemist Ex Machina: Advanced Synthesis Planning by Computers. *Acc. Chem. Res.* **2021**, *54* (5), 1094–1106.

(17) Dou, B.; Zhu, Z.; Merkurjev, E.; Ke, L.; Chen, L.; Jiang, J.; Zhu, Y.; Liu, J.; Zhang, B.; Wei, G. W. Machine Learning Methods for Small Data Challenges in Molecular Science. *Chem. Rev.* **2023**, *123* (13), 8736–8780.

(18) Nandy, A.; Duan, C.; Taylor, M. G.; Liu, F.; Steeves, A. H.; Kulik, H. J. Computational Discovery of Transition-metal Complexes: From High-throughput Screening to Machine Learning. *Chem. Rev.* **2021**, *121* (16), 9927–10000.

(19) Friederich, P.; Hase, F.; Proppe, J.; Aspuru-Guzik, A. Machine-learned potentials for next-generation matter simulations. *Nat. Mater.* **2021**, *20* (6), 750–761.

(20) Mai, H.; Le, T. C.; Chen, D.; Winkler, D. A.; Caruso, R. A. Machine Learning for Electrocatalyst and Photocatalyst Design and Discovery. *Chem. Rev.* **2022**, *122* (16), 13478–13515.

(21) Townsend, J.; Micucci, C. P.; Hymel, J. H.; Maroulas, V.; Vogiatzis, K. D. Representation of molecular structures with persistent homology for machine learning applications in chemistry. *Nat. Commun.* **2020**, *11* (1), 3230.

(22) Jones, G. M.; Smith, B. A.; Kirkland, J. K.; Vogiatzis, K. D. Data-driven ligand field exploration of Fe(IV)–oxo sites for C–H activation. *Inorg. Chem. Front.* **2023**, *10* (4), 1062–1075.

(23) Noe, F.; Tkatchenko, A.; Muller, K. R.; Clementi, C. Machine Learning for Molecular Simulation. *Annu. Rev. Phys. Chem.* **2020**, *71*, 361–390.

(24) Zubatiuk, T.; Isayev, O. Development of Multimodal Machine Learning Potentials: Toward a Physics-Aware Artificial Intelligence. *Acc. Chem. Res.* **2021**, *54* (7), 1575–1585.

(25) Behler, J. Four Generations of High-Dimensional Neural Network Potentials. *Chem. Rev.* **2021**, *121* (16), 10037–10072.

(26) Musil, F.; Grisafi, A.; Bartok, A. P.; Ortner, C.; Csanyi, G.; Ceriotti, M. Physics-Inspired Structural Representations for Molecules and Materials. *Chem. Rev.* **2021**, *121* (16), 9759–9815.

(27) Huang, B.; von Lilienfeld, O. A. Ab Initio Machine Learning in Chemical Compound Space. *Chem. Rev.* **2021**, *121* (16), 10001–10036.

(28) Keith, J. A.; Vassilev-Galindo, V.; Cheng, B.; Chmiela, S.; Gastegger, M.; Muller, K. R.; Tkatchenko, A. Combining Machine Learning and Computational Chemistry for Predictive Insights Into Chemical Systems. *Chem. Rev.* **2021**, *121* (16), 9816–9872.

(29) Queen, O.; McCarver, G. A.; Thatigotla, S.; Abolins, B. P.; Brown, C. L.; Maroulas, V.; Vogiatzis, K. D. Polymer graph neural networks for multitask property learning. *npj Comput. Mater.* **2023**, *9* (1), No. 90.

(30) Aldossary, A.; Campos-Gonzalez-Angulo, J. A.; Pablo-Garcia, S.; Leong, S. X.; Rajaonson, E. M.; Thiede, L.; Tom, G.; Wang, A.; Avagliano, D.; Aspuru-Guzik, A. In Silico Chemical Experiments in the Age of AI: From Quantum Chemistry to Machine Learning and Back. *Adv. Mater.* **2024**, *36* (30), No. 2402369.

(31) Dral, P. O. Quantum Chemistry in the Age of Machine Learning. *J. Phys. Chem. Lett.* **2020**, *11* (6), 2336–2347.

(32) Tew, D. P.; Klopper, W.; Helgaker, T. Electron correlation: the many-body problem at the heart of chemistry. *J. Comput. Chem.* **2007**, *28* (8), 1307–1320.

(33) Townsend, J.; Kirkland, J. K.; Vogiatzis, K. D. Post-Hartree-Fock methods: configuration interaction, many-body perturbation theory, coupled-cluster theory. In *Mathematical Physics in Theoretical Chemistry*; Blinder, S. M., House, J. E., Eds.; Elsevier, 2019; pp 63–117.

(34) Campbell, C.; Sands, S.; Ferraro, C.; Tsao, H.-Y.; Mavrommatis, A. From data to action: How marketers can leverage AI. *Bus. Horiz.* **2020**, *63* (2), 227–243.

(35) Ferraro, C.; Demsar, V.; Sands, S.; Restrepo, M.; Campbell, C. The paradoxes of generative AI-enabled customer service: A guide for managers. *Bus. Horiz.* **2024**, *67* (5), 549–559.

(36) Cui, Y.; van Esch, P.; Phelan, S. How to build a competitive advantage for your brand using generative AI. *Bus. Horiz.* **2024**, *67* (5), 583–594.

(37) Mowbray, M.; Vallerio, M.; Perez-Galvan, C.; Zhang, D.; Del Rio Chanona, A.; Navarro-Brull, F. J. Industrial data science – a review of machine learning applications for chemical and process industries. *React. Chem. Eng.* **2022**, *7* (7), 1471–1509.

(38) Coe, J. P. Machine Learning Configuration Interaction for ab Initio Potential Energy Curves. *J. Chem. Theory Comput.* **2019**, *15* (11), 6179–6189.

(39) Goings, J. J.; Hu, H.; Yang, C.; Li, X. Reinforcement Learning Configuration Interaction. *J. Chem. Theory Comput.* **2021**, *17* (9), 5482–5491.

(40) Yang, P.-J.; Sugiyama, M.; Tsuda, K.; Yanai, T. Artificial Neural Networks Applied as Molecular Wave Function Solvers. *J. Chem. Theory Comput.* **2020**, *16* (6), 3513–3529.

(41) Pfau, D.; Spencer, J. S.; Matthews, A. G. D. G.; Foulkes, W. M. C. Ab initio solution of the many-electron Schrödinger equation with deep neural networks. *Phys. Rev. Res.* **2020**, *2*, No. 033429.

(42) Glehn, v. S.; J, S.; Pfau, D. A Self-Attention Ansatz for Ab-Initio Quantum Chemistry. *arXiv* **2022**, arXiv:2211.13672.

(43) Hermann, J.; Schätzle, Z.; Noé, F. Deep-neural-network solution of the electronic Schrödinger equation. *Nat. Chem.* **2020**, *12*, 891–897.

(44) Sumanta, K.; Ghosh, D. G. Machine learning matrix product state ansatz for strongly correlated systems. *J. Chem. Phys.* **2023**, *158* (6), No. 064108.

(45) Ramakrishnan, R.; Dral, P. O.; Rupp, M.; von Lilienfeld, O. A. Big Data Meets Quantum Chemistry Approximations: The Δ -Machine Learning Approach. *J. Chem. Theory Comput.* **2015**, *11* (5), 2087–2096.

(46) Schütt, K. T.; Gastegger, M.; Tkatchenko, A.; Müller, K.-R.; Maurer, R. J. Unifying machine learning and quantum chemistry with a deep neural network for molecular wavefunctions. *Nat. Commun.* **2019**, *10*, 5024.

(47) Cheng, L.; Kovachki, N. B.; Welborn, M.; Miller, T. F., III Regression Clustering for Improved Accuracy and Training Costs with Molecular-Orbital-Based Machine Learning. *J. Chem. Theory Comput.* **2019**, *15* (12), 6668–6677.

(48) Cheng, L.; Sun, J.; Miller, T. F., III Accurate Molecular-Orbital-Based Machine Learning Energies via Unsupervised Clustering of Chemical Space. *J. Chem. Theory Comput.* **2022**, *18* (8), 4826–4835.

(49) Welborn, M.; Cheng, L.; Miller, T. F., III Transferability in Machine Learning for Electronic Structure via the Molecular Orbital Basis. *J. Chem. Theory Comput.* **2018**, *14* (9), 4772–4779.

(50) Cheng, L.; Welborn, M.; Christensen, A. S.; Miller, T. F., III A universal density matrix functional from molecular orbital-based machine learning: Transferability across organic molecules. *J. Chem. Phys.* **2019**, *150* (13), No. 131103.

(51) Qiao, Z.; Christensen, A. S.; Welborn, M.; Manby, F. R.; Anandkumar, A.; Miller, T. F., III Informing geometric deep learning with electronic interactions to accelerate quantum chemistry. *Proc. Natl. Acad. Sci. U. S. A.* **2022**, *119* (31), No. e2205221119.

(52) Townsend, J.; Vogiatzis, K. D. Data-Driven Acceleration of the Coupled-Cluster Singles and Doubles Iterative Solver. *J. Phys. Chem. Lett.* **2019**, *10* (14), 4129–4135.

- (53) Wigh, D. S.; Goodman, J. M.; Lapkin, A. A. A review of molecular representation in the age of machine learning. *WIREs Comput. Mol. Sci.* **2022**, *12* (5), No. e1603.
- (54) Raghunathan, S.; Priyakumar, U. D. Molecular representations for machine learning applications in chemistry. *Int. J. Quantum Chem.* **2022**, *122* (7), No. e26870.
- (55) Jones, G. M.; Story, B.; Maroulas, V.; Vogiatzis, K. D. *Molecular Representations for Machine Learning*; American Chemical Society, ACS In Focus, 2023.
- (56) Schiff, Y.; Chenthamarakshan, V.; Hoffman, S. C.; Ramamurthy, K. N.; Das, P. Augmenting Molecular Deep Generative Models with Topological Data Analysis Representations. In *ICASSP 2022–2022 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Singapore, **2022**.
- (57) Schütt, K. T.; Arbabzadah, F.; Chmiela, S.; Müller, K. R.; Tkatchenko, A. Quantum-chemical insights from deep tensor neural networks. *Nat. Commun.* **2017**, *8*, No. 13890.
- (58) Meyer, B.; Sawatlon, B.; Heinen, S.; von Lilienfeld, O. A.; Corminboeuf, C. Machine learning meets volcano plots: computational discovery of cross-coupling catalysts. *Chem. Sci.* **2018**, *9* (35), 7069–7077.
- (59) Dotson, J. J.; van Dijk, L.; Timmerman, J. C.; Grosslight, S.; Walroth, R. C.; Gosselin, F.; Puntener, K.; Mack, K. A.; Sigman, M. S. Data-Driven Multi-Objective Optimization Tactics for Catalytic Asymmetric Reactions Using Bisphosphine Ligands. *J. Am. Chem. Soc.* **2023**, *145* (1), 110–121.
- (60) Vargas, S.; Hennefarth, M. R.; Liu, Z.; Alexandrova, A. N. Machine Learning to Predict Diels-Alder Reaction Barriers from the Reactant State Electron Density. *J. Chem. Theory Comput.* **2021**, *17* (10), 6203–6213.
- (61) Zhang, Z. J.; Li, S. W.; Oliveira, J. C. A.; Li, Y.; Chen, X.; Zhang, S. Q.; Xu, L. C.; Rogge, T.; Hong, X.; Ackermann, L. Data-driven design of new chiral carboxylic acid for construction of indoles with C-central and C-N axial chirality via cobalt catalysis. *Nat. Commun.* **2023**, *14* (1), 3149.
- (62) Vennelakanti, V.; Kilic, I. B.; Terrones, G. G.; Duan, C.; Kulik, H. J. Machine Learning Prediction of the Experimental Transition Temperature of Fe(II) Spin-Crossover Complexes. *J. Phys. Chem. A* **2024**, *128* (1), 204–216.
- (63) Zhang, Z.; Schott, J. A.; Liu, M.; Chen, H.; Lu, X.; Sumpter, B. G.; Fu, J.; Dai, S. Prediction of Carbon Dioxide Adsorption via Deep Learning. *Angew. Chem., Int. Ed.* **2019**, *58* (1), 259–263.
- (64) Gokcan, H.; Isayev, O. Learning molecular potentials with neural networks. *WIREs Comp. Mol. Sci.* **2022**, *12* (2), No. e1564.
- (65) Li, J.; Lopez, S. A. A Look Inside the Black Box of Machine Learning Photodynamics Simulations. *Acc. Chem. Res.* **2022**, *55* (14), 1972–1984.
- (66) Nesbet, R. K. Electronic Correlation in Atoms and Molecules. *Advances in Chemical Physics* **1965**, *9*, 321–363.
- (67) Van Voorhis, T.; Head-Gordon, M. Benchmark variational coupled cluster doubles results. *J. Chem. Phys.* **2000**, *113* (20), 8873–8879.
- (68) Evangelista, F. A. Alternative single-reference coupled cluster approaches for multireference problems: the simpler, the better. *J. Chem. Phys.* **2011**, *134* (22), No. 224102.
- (69) Marie, A.; Kossoski, F.; Loos, P. F. Variational coupled cluster for ground and excited states. *J. Chem. Phys.* **2021**, *155* (10), No. 104105.
- (70) Jones, G. M.; Pathirage, P. D. V. S.; Vogiatzis, K. D. Data-driven acceleration of coupled-cluster and perturbation theory methods. In *Quantum Chemistry in the Age of Machine Learning*; Dral, P. O., Ed.; Elsevier: 2023.
- (71) Pathirage, P. D. V. S.; Phillips, J. T.; Vogiatzis, K. D. Exploration of the Two-Electron Excitation Space with Data-Driven Coupled Cluster. *J. Phys. Chem. A* **2024**, *128* (10), 1938–1947.
- (72) Smith, D. G. A.; Burns, L. A.; Sirianni, D. A.; Nascimento, D. R.; Kumar, A.; James, A. M.; Schriber, J. B.; Zhang, T.; Zhang, B.; Abbott, A. S.; et al. Psi4NumPy: An Interactive Quantum Chemistry Programming Environment for Reference Implementations and Rapid Development. *J. Chem. Theory Comput.* **2018**, *14* (7), 3504–3511.
- (73) Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; et al. Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.* **2011**, *12*, 2825–2830.
- (74) Townsend, J.; Vogiatzis, K. D. Transferable MP2-Based Machine Learning for Accurate Coupled-Cluster Energies. *J. Chem. Theory Comput.* **2020**, *16* (12), 7453–7461.
- (75) Ng, W.-P.; Liang, Q.; Yang, J. Low-Data Deep Quantum Chemical Learning for Accurate MP2 and Coupled-Cluster Correlations. *J. Chem. Theory Comput.* **2023**, *19* (16), 5439–5449.
- (76) Schrader, S. E.; Kvaal, S. Accelerated coupled cluster calculations with Procrustes orbital interpolation. *J. Chem. Phys.* **2023**, *158* (11), No. 114116.
- (77) Peyton, B. G.; Briggs, C.; D’Cunha, R.; Margraf, J. T.; Crawford, T. D. Machine-Learning Coupled Cluster Properties through a Density Tensor Representation. *J. Phys. Chem. A* **2020**, *124* (23), 4861–4871.
- (78) Ruth, M.; Gerbig, D.; Schreiner, P. R. Machine Learning of Coupled Cluster (T)-Energy Corrections via Delta (Delta)-Learning. *J. Chem. Theory Comput.* **2022**, *18* (8), 4846–4855.
- (79) Jones, G. M.; Li, R. R.; DePrince, A. E., 3rd; Vogiatzis, K. D. Data-Driven Refinement of Electronic Energies from Two-Electron Reduced-Density-Matrix Theory. *J. Phys. Chem. Lett.* **2023**, *14* (28), 6377–6385.
- (80) Raissi, M.; Perdikaris, P.; Karniadakis, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* **2019**, *378*, 686–707.
- (81) Raghavachari, K.; Trucks, G. W.; Pople, J. A.; Head-Gordon, M. A fifth-order perturbation comparison of electron correlation theories. *Chem. Phys. Lett.* **1989**, *157* (6), 479–483.